

Binary Numbers

Binary Numbers

Human beings do arithmetic using the decimal (base 10) number system. Most computers do arithmetic using the binary (base 2) number system. It may seem otherwise, since communication with the computer (input/output) is in base 10 numbers. This transparency does not mean that the computer uses base 10. In fact, it converts inputs to base 2 (or perhaps base 16), then performs base 2 arithmetic, and finally, translates the answer into base 10 before it displays a result. Some experimentation is required to verify this. One computer with nine decimal digits of accuracy gave the answer

$$(1) \quad \sum_{k=1}^{100,000} 0.1 = 9999.99447.$$

Here the intent was to add the number $\frac{1}{10}$ repeatedly 100,000 times. The mathematical answer is exactly 10,000. One goal is to understand the reason for the computer's apparently flawed calculation. At the end of this section it will be shown how something is lost when the computer translates the decimal fraction $\frac{1}{10}$ into a binary number.

Base 2 Numbers

Base 10 numbers are used for most mathematical purposes. For illustration, the number 1563 is expressible in *expanded form* as

$$1563 = (1 \times 10^3) + (5 \times 10^2) + (6 \times 10^1) + (3 \times 10^0).$$

In general, let N denote a positive integer; then the digits $a_0, a_1, \dots, a_k$ exist so that N has the base 10 expansion

$$N = (a_k \times 10^k) + (a_{k-1} \times 10^{k-1}) + \dots + (a_1 \times 10^1) + (a_0 \times 10^0),$$

where the digits a_k are chosen from $\{0, 1, \dots, 8, 9\}$. Thus N is expressed in decimal notation as

$$(2) \quad N = a_k a_{k-1} \dots a_2 a_1 a_0_{\text{ten}} \quad (\text{decimal}).$$

If it is understood that 10 is the base, then (2) is written as

$$N = a_k a_{k-1} \dots a_2 a_1 a_0.$$

For example, we understand that $1563 = 1563_{\text{ten}}$.

Using powers of 2, the number 1563 can be written

$$(3) \quad \begin{aligned} 1563 &= (1 \times 2^{10}) + (1 \times 2^9) + (0 \times 2^8) + (0 \times 2^7) + (0 \times 2^6) \\ &\quad + (0 \times 2^5) + (1 \times 2^4) + (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) \\ &\quad + (1 \times 2^0). \end{aligned}$$

This can be verified by performing the calculation

$$1563 = 1024 + 512 + 16 + 8 + 2 + 1.$$

In general, let N denote a positive integer; the digits $b_0, b_1, \dots, b_J$ exist so that N has the base 2 expansion

$$(4) \quad N = (b_J \times 2^J) + (b_{J-1} \times 2^{J-1}) + \dots + (b_1 \times 2^1) + (b_0 \times 2^0),$$

where each digit b_j is either a 0 or 1. Thus N is expressed in binary notation as

$$(5) \quad N = b_J b_{J-1} \dots b_2 b_1 b_0_{\text{two}} \quad (\text{binary}).$$

Using the notation (5) and the result in (3) yields

$$1563 = 11000011011_{\text{two}}.$$

Remarks. The word “two” will always be used as a subscript at the end of a binary number. This will enable the reader to distinguish binary numbers from the ordinary base 10 usage. Thus 111 means one hundred eleven, whereas 111_{two} stands for seven.

Example 1.10. Show how to obtain $1563 = 11000011011_{\text{two}}$.

Start with $N = 1563$ and construct the quotients and remainders according to the equations in (8):

$$\begin{aligned} 1563 &= 2 \times 781 + 1, & b_0 &= 1 \\ 781 &= 2 \times 390 + 1, & b_1 &= 1 \\ 390 &= 2 \times 195 + 0, & b_2 &= 0 \\ 195 &= 2 \times 97 + 1, & b_3 &= 1 \\ 97 &= 2 \times 48 + 1, & b_4 &= 1 \\ 48 &= 2 \times 24 + 0, & b_5 &= 0 \\ 24 &= 2 \times 12 + 0, & b_6 &= 0 \\ 12 &= 2 \times 6 + 0, & b_7 &= 0 \\ 6 &= 2 \times 3 + 0, & b_8 &= 0 \\ 3 &= 2 \times 1 + 1, & b_9 &= 1 \\ 1 &= 2 \times 0 + 1, & b_{10} &= 1. \end{aligned}$$

Sequences and Series

When rational numbers are expressed in decimal form, it is often the case that infinitely many digits are required. A familiar example is

$$(9) \quad \frac{1}{3} = 0.\overline{3}.$$

Here the symbol $\overline{3}$ means that the digit 3 is repeated forever to form an infinite repeating decimal. It is understood that 10 is the base in (9). Moreover, it is the mathematical intent that (9) is the shorthand notation for the infinite series

$$(10) \quad \begin{aligned} S &= (3 \times 10^{-1}) + (3 \times 10^{-2}) + \cdots + (3 \times 10^{-n}) + \cdots \\ &= \sum_{k=1}^{\infty} 3(10)^{-k} = \frac{1}{3}. \end{aligned}$$

If only a finite number of digits is displayed, then an approximation to $1/3$ is obtained. For example, $1/3 \approx 0.333 = 333/1000$. The error in this approximation is $1/3000$. Using (10), the reader can verify that $1/3 = 0.333 + 1/3000$.

It is important to understand the expansion in (10). A naive approach is to multiply both sides by 10 and then subtract.

Definition 1.6. The infinite series

$$(11) \quad \sum_{n=0}^{\infty} cr^n = c + cr + cr^2 + \cdots + cr^n + \cdots ,$$

where $c \neq 0$ and $r \neq 0$, is called a *geometric series* with ratio r .

Theorem 1.14 (Geometric Series). The geometric series has the following properties:

$$(12) \quad \text{If } |r| < 1, \text{ then } \sum_{n=0}^{\infty} cr^n = \frac{c}{1-r}.$$

$$(13) \quad \text{If } |r| > 1, \text{ then the series diverges.}$$

Proof. The summation formula for a finite geometric series is

$$(14) \quad S_n = c + cr + cr^2 + \cdots + cr^n = \frac{c(1 - r^{n+1})}{1 - r} \quad \text{for } r \neq 1.$$

To establish (12), observe that

$$(15) \quad |r| < 1 \quad \text{implies that} \quad \lim_{n \rightarrow \infty} r^{n+1} = 0.$$

Taking the limit as $n \rightarrow \infty$, use (14) and (15) to get

$$\lim_{n \rightarrow \infty} S_n = \frac{c}{1-r} \left(1 - \lim_{n \rightarrow \infty} r^{n+1} \right) = \frac{c}{1-r}.$$

By equation (15) of Section 1.1, the limit above establishes (12).

When $|r| \geq 1$, the sequence $\{r^{n+1}\}$ does not converge. Hence the sequence $\{S_n\}$ in (14) does not tend to a limit. Therefore, (13) is established. •

Binary Fractions

Binary (base 2) fractions can be expressed as sums involving negative powers of 2. If R is a real number that lies in the range $0 < R < 1$, there exist digits $d_1, d_2, \dots, d_n, \dots$ so that

$$(16) \quad R = (d_1 \times 2^{-1}) + (d_2 \times 2^{-2}) + \cdots + (d_n \times 2^{-n}) + \cdots,$$

where $d_j \in \{0, 1\}$. We usually express the quantity on the right side of (16) in the binary fraction notation

$$(17) \quad R = 0.d_1d_2 \cdots d_n \cdots_{\text{two}}.$$

$$\frac{7}{10} = 0.\overline{10110}_{\text{two}}.$$

Example 1.12. The binary decimal representation of $7/10$ given in (18) was found using the formulas in (22). Let $R = 7/10 = 0.7$; then

$2R = 1.4$	$d_1 = \text{int}(1.4) = 1$	$F_1 = \text{frac}(1.4) = 0.4$
$2F_1 = 0.8$	$d_2 = \text{int}(0.8) = 0$	$F_2 = \text{frac}(0.8) = 0.8$
$2F_2 = 1.6$	$d_3 = \text{int}(1.6) = 1$	$F_3 = \text{frac}(1.6) = 0.6$
$2F_3 = 1.2$	$d_4 = \text{int}(1.2) = 1$	$F_4 = \text{frac}(1.2) = 0.2$
$2F_4 = 0.4$	$d_5 = \text{int}(0.4) = 0$	$F_5 = \text{frac}(0.4) = 0.4$
$2F_5 = 0.8$	$d_6 = \text{int}(0.8) = 0$	$F_6 = \text{frac}(0.8) = 0.8$
$2F_6 = 1.6$	$d_7 = \text{int}(1.6) = 1$	$F_7 = \text{frac}(1.6) = 0.6$

Note that $2F_2 = 1.6 = 2F_6$. The patterns $d_k = d_{k+4}$ and $F_k = F_{k+4}$ will occur for $k = 2, 3, 4, \dots$. Thus $7/10 = 0.10110_{\text{two}}$. ■

Geometric series can be used to find the base 10 rational number that a binary number represents.

Example 1.13. Find the base 10 rational number that the binary number $0.\overline{01}_{\text{two}}$ represents. In expanded form,

$$\begin{aligned}
 0.\overline{01}_{\text{two}} &= (0 \times 2^{-1}) + (1 \times 2^{-2}) + (0 \times 2^{-3}) + (1 \times 2^{-4}) + \dots \\
 &= \sum_{k=1}^{\infty} (2^{-2})^k = -1 + \sum_{k=0}^{\infty} (2^{-2})^k \\
 &= -1 + \frac{1}{1 - \frac{1}{4}} = -1 + \frac{4}{3} = \frac{1}{3}.
 \end{aligned}$$

Machine Numbers

Computers use a normalized floating-point binary representation for real numbers. This means that the mathematical quantity x is not actually stored in the computer. Instead, the computer stores a binary approximation to x :

$$(26) \quad x \approx \pm q \times 2^n.$$

The number q is the *mantissa* and it is a finite binary expression satisfying the inequality $1/2 \leq q < 1$. The integer n is called the *exponent*.

In a computer, only a small subset of the real number system is used. Typically, this subset contains only a portion of the binary numbers suggested by (26). The number of binary digits is restricted in both the numbers q and n . For example, consider the set of all positive real numbers of the form

$$(27) \quad 0.\underline{d_1 d_2 d_3 d_4}_{\text{two}} \times 2^n,$$

where $d_1 = 1$ and d_2, d_3 , and d_4 are either 0 or 1, and $n \in \{-3, -2, -1, 0, 1, 2, 3, 4\}$. There are eight choices for the mantissa and eight choices for the exponent in (27), and this produces a set of 64 numbers:

$$(28) \quad \{0.1000_{\text{two}} \times 2^{-3}, 0.1001_{\text{two}} \times 2^{-3}, \dots, 0.1110_{\text{two}} \times 2^4, 0.1111_{\text{two}} \times 2^4\}.$$

Computer Floating-Point Numbers

Computers have both an *integer mode* and a *floating-point mode* for representing numbers. The integer mode is used for performing calculations that are known to be integer valued and has limited usage for numerical analysis. Floating-point numbers are used for scientific and engineering applications. It must be understood that any computer implementation of equation (26) places restrictions on the number of digits used in the mantissa q , and that the range of possible exponents n must be limited.

Computers that use 32 bits to represent single-precision real numbers use 8 bits for the exponent and 24 bits for the mantissa. They can represent real numbers with magnitudes in the range

$$2.938736E - 39 \quad \text{to} \quad 1.701412E + 38$$

(i.e., 2^{-128} to 2^{127}) with six decimal digits of numerical precision (e.g., $2^{-23} = 1.2 \times 10^{-7}$).

Computers that use 48 bits to represent single-precision real numbers might use 8 bits for the exponent and 40 bits for the mantissa. They can represent real numbers in the range

$$2.9387358771E - 39 \quad \text{to} \quad 1.7014118346E + 38$$

(i.e., 2^{-128} to 2^{127}) with 11 decimal digits of numerical precision (e.g., $2^{-39} = 1.8 \times 10^{-12}$).

If the computer has 64-bit double-precision real numbers, it might use 11 bits for the exponent and 53 bits for the mantissa. They can represent real numbers in the range

$$5.562684646268003E - 309 \quad \text{to} \quad 8.988465674311580E + 307$$

(i.e., 2^{-1024} to 2^{1023}) with about 16 decimal digits of numerical precision (e.g., $2^{-52} = 2.2 \times 10^{-16}$).

1.3 Error Analysis

In the practice of numerical analysis it is important to be aware that computed solutions are not exact mathematical solutions. The precision of a numerical solution can be diminished in several subtle ways. Understanding these difficulties can often guide the practitioner in the proper implementation and/or development of numerical algorithms.

Definition 1.7. Suppose that $\hat{p}$ is an approximation to p . The absolute error is $E_p = |p - \hat{p}|$, and the relative error is $R_p = |p - \hat{p}|/|p|$, provided that $p \neq 0$. ▲

Definition 1.8. The number $\hat{p}$ is said to *approximate* p to d significant digits if d is the largest nonnegative integer for which

$$(2) \quad \frac{|p - \hat{p}|}{|p|} < \frac{10^{1-d}}{2}.$$

Example 1.15. Determine the number of significant digits for the approximations in Example 1.14.

- (3a) If $x = 3.141592$ and $\hat{x} = 3.14$, then $|x - \hat{x}|/|x| = 0.000507 < 10^{-2}/2$. Therefore, $\hat{x}$ approximates x to two significant digits.
- (3b) If $y = 1,000,000$ and $\hat{y} = 999,996$, then $|y - \hat{y}|/|y| = 0.000004 < 10^{-5}/2$. Therefore, $\hat{y}$ approximates y to six significant digits.
- (3c) If $z = 0.000012$ and $\hat{z} = 0.000009$, then $|z - \hat{z}|/|z| = 0.25 < 10^{-0}/2$. Therefore, $\hat{z}$ approximates z to one significant digit. ■

1. Absolute vs. Relative Error Bounds (for $\sqrt{2}$)

When analyzing approximations, we use two metrics to quantify error. Let's use the **chopped approximation** of $\sqrt{2}$ to 7 significant digits:

- **True Value (x):** 1.41421356237...
- **Approximated Value ($\hat{x}$):** 1.414213

Absolute Error

- **Definition:** The exact magnitude of the discrepancy.
- **Formula:** Absolute Error = $|x - \hat{x}|$
- **Calculation:** $|1.41421356237 - 1.414213| = 0.00000056237$
- **Limitation:** It does not show how severe the error is relative to the size of the number.

Relative Error

- **Definition:** The error scaled against the true value. This is critical for assessing precision.
- **Formula:** Relative Error = $\frac{|x - \hat{x}|}{|x|}$
- **Calculation:** $\frac{0.00000056237}{1.41421356237} \approx 0.00000039768$ (or 0.000039768%)
- **Bound:** For a d -digit chopped decimal floating-point number, the relative error bound is always limited by machine epsilon ($\epsilon_m = 10^{1-d}$). Here, $10^{1-7} = 10^{-6}$. Our calculated relative error (3.97×10^{-7}) safely falls under this maximum bound.

Truncation Error

The notion of truncation error usually refers to errors introduced when a more complicated mathematical expression is “replaced” with a more elementary formula. This terminology originates from the technique of replacing a complicated function with a truncated Taylor series. For example, the infinite Taylor series

$$e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \frac{x^8}{4!} + \cdots + \frac{x^{2n}}{n!} + \cdots$$

might be replaced with just the first five terms $1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \frac{x^8}{4!}$. This might be done when approximating an integral numerically.

Example 1.16. Given that $\int_0^{1/2} e^{x^2} dx = 0.544987104184 = p$, determine the accuracy of the approximation obtained by replacing the integrand $f(x) = e^{x^2}$ with the truncated Taylor series $P_8(x) = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \frac{x^8}{4!}$.
Term-by-term integration produces

$$\begin{aligned} \int_0^{1/2} \left(1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \frac{x^8}{4!} \right) dx &= \left(x + \frac{x^3}{3} + \frac{x^5}{5(2!)} + \frac{x^7}{7(3!)} + \frac{x^9}{9(4!)} \right) \Big|_{x=0}^{x=1/2} \\ &= \frac{1}{2} + \frac{1}{24} + \frac{1}{320} + \frac{1}{5376} + \frac{1}{110,592} \\ &= \frac{2,109,491}{3,870,720} = 0.544986720817 = \hat{p}. \end{aligned}$$

Since

$$10^{-5}/2 > |p - \hat{p}|/|p| = 7.03442 \times 10^{-7} > 10^{-6}/2,$$

the approximation $\hat{p}$ agrees with the true answer $p = 0.544987104184$ to five significant digits. The graphs of $f(x) = e^{x^2}$ and $y = P_8(x)$ and the area under the curve for $0 \leq x \leq 1/2$ are shown in Figure 1.7. ■

Chopping Off versus Rounding Off

Consider any real number p that is expressed in *normalized decimal form*:

$$(4) \quad p = \pm 0.d_1 d_2 d_3 \cdots d_k d_{k+1} \cdots \times 10^n,$$

where $1 \leq d_1 \leq 9$ and $0 \leq d_j \leq 9$ for $j > 1$. Suppose that k is the maximum number of decimal digits carried in the floating-point computations of a computer; then the real number p is represented by $fl_{\text{chop}}(p)$, which is given by

$$(5) \quad fl_{\text{chop}}(p) = \pm 0.d_1 d_2 d_3 \cdots d_k \times 10^n,$$

where $1 \leq d_1 \leq 9$ and $0 \leq d_j \leq 9$ for $1 < j \leq k$. The number $fl_{\text{chop}}(p)$ is called the *chopped floating-point representation* of p . In this case the k th digit of $fl_{\text{chop}}(p)$ agrees with the k th digit of p . An alternative k -digit representation is the *rounded floating-point representation* $fl_{\text{round}}(p)$, which is given by

$$(6) \quad fl_{\text{round}}(p) = \pm 0.d_1 d_2 d_3 \cdots r_k \times 10^n,$$

where $1 \leq d_1 \leq 9$ and $0 \leq d_j \leq 9$ for $1 < j < k$ and the last digit, r_k , is obtained by rounding the number $d_k d_{k+1} d_{k+2} \dots$ to the nearest integer. For example, the real number

$$p = \frac{22}{7} = 3.142857142857142857 \dots$$

has the following six-digit representations:

$$fl_{\text{chop}}(p) = 0.314285 \times 10^1,$$

$$fl_{\text{round}}(p) = 0.314286 \times 10^1.$$

For common purposes the chopping and rounding would be written as 3.14285 and 3.14286, respectively. The reader should note that essentially all computers use some form of the rounded floating-point representation method.

1. Truncation Error (Algorithmic)

This occurs when you stop a mathematical algorithm early.

- **Method:** Finding $\sqrt{2}$ using Newton's iterative method.
- **Process:** Starting at $x_0 = 1$, the next step yields 1.5.
- **Early Stop:** Stopping at the second step gives **1.41666666...**
- **The Error:** The difference ($1.416666 - 1.414213$) leaves a truncation error of **0.00245310**.

2. Chopping Error (Storage)

This occurs when a system discards extra digits without rounding.

- **Constraint:** A computer storage system limited to **7 significant digits**.
- **Action:** The system drops everything after the 7th digit.
- **Result:** 1.41421356... becomes **1.414213**.
- **The Error:** Discarding the remaining digits leaves a chopping error of **0.00000056**

3. Round-Off Error (Storage)

This occurs when a system rounds up or down to the nearest value.

- **Constraint:** A computer storage system limited to **7 significant digits**.
- **Action:** The system looks at the 8th digit (5).
- **Result:** It rounds the 7th digit up to **1.414214**.
- **The Error:** The adjustments leave a small round-off error of **-0.00000044**.

1. **Multiplication Error:** Multiplying two numbers with round-off errors expands the total error bound linearly.
2. **Addition (Accumulation) Error:** Summing n terms forces the computer to continuously align exponents. Tiny fractional parts are repeatedly chopped or rounded off.
3. **Worst-Case Growth Bound:** The theoretical error bound for standard matrix multiplication scales by $\mathcal{O}(n \cdot \epsilon_m)$. If you multiply two large 1000×1000 matrices, the baseline round-off error can magnify by a factor of 1,000.

$\mathcal{O}(h^n)$ Order of Approximation

Clearly, the sequences $\left\{\frac{1}{n^2}\right\}_{n=1}^{\infty}$ and $\left\{\frac{1}{n}\right\}_{n=1}^{\infty}$ are both converging to zero. In addition, it should be observed that the first sequence is converging to zero more rapidly than the second sequence. In the coming chapters some special terminology and notation will be used to describe how rapidly a sequence is converging.

Definition 1.9. The function $f(h)$ is said to be **big Oh of $g(h)$** , denoted $f(h) = \mathcal{O}(g(h))$, if there exist constants C and c such that

$$(7) \quad |f(h)| \leq C|g(h)| \quad \text{whenever } h \geq c. \quad \blacktriangle$$

Example 1.20. Consider the functions $f(x) = x^2 + 1$ and $g(x) = x^3$. Since $x^2 \leq x^3$ and $1 \leq x^3$ for $x \geq 1$, it follows that $x^2 + 1 \leq 2x^3$ for $x \geq 1$. Therefore, $f(x) = \mathcal{O}(g(x))$. ■

The big Oh notation provides a **useful way of describing the rate of growth of a function in terms of well-known elementary functions** (x^n , $x^{1/n}$, a^x , $\log_a x$, etc.).

The **rate of convergence of sequences** can be described in a similar manner.

Definition 1.10. Let $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ be two sequences. The sequence $\{x_n\}$ is said to be of order big Oh of $\{y_n\}$, denoted $x_n = \mathcal{O}(y_n)$, if there exist constants C and N such that

$$(8) \quad |x_n| \leq C|y_n| \quad \text{whenever } n \geq N. \quad \blacktriangle$$

Example 1.21. $\frac{n^2 - 1}{n^3} = \mathcal{O}\left(\frac{1}{n}\right)$, since $\frac{n^2 - 1}{n^3} \leq \frac{n^2}{n^3} = \frac{1}{n}$ whenever $n \geq 1$. ■

Often a function $f(h)$ is approximated by a function $p(h)$ and the error bound is known to be $M|h^n|$. This leads to the following definition.

Definition 1.11. Assume that $f(h)$ is approximated by the function $p(h)$ and that there exist a real constant $M > 0$ and a positive integer n so that

$$(9) \quad \frac{|f(h) - p(h)|}{|h^n|} \leq M \quad \text{for sufficiently small } h.$$

We say that $p(h)$ **approximates** $f(h)$ with order of approximation $\mathcal{O}(h^n)$ and write

$$(10) \quad f(h) = p(h) + \mathcal{O}(h^n). \quad \blacktriangle$$

When relation (9) is rewritten in the form $|f(h) - p(h)| \leq M|h^n|$, we see that the notation $O(h^n)$ stands in place of the error bound $M|h^n|$. The following results show how to apply the definition to simple combinations of two functions.

Theorem 1.15. Assume that $f(h) = p(h) + O(h^n)$, $g(h) = q(h) + O(h^m)$, and $r = \min\{m, n\}$. Then

$$(11) \quad f(h) + g(h) = p(h) + q(h) + O(h^r),$$

$$(12) \quad f(h)g(h) = p(h)q(h) + O(h^r),$$

and

$$(13) \quad \frac{f(h)}{g(h)} = \frac{p(h)}{q(h)} + O(h^r) \quad \text{provided that } g(h) \neq 0 \text{ and } q(h) \neq 0.$$

Theorem 1.16 (Taylor's Theorem). Assume that $f \in C^{n+1}[a, b]$. If both x_0 and $x = x_0 + h$ lie in $[a, b]$, then

$$(15) \quad f(x_0 + h) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} h^k + O(h^{n+1}).$$

K+1st deg term neglected

Kth degree

The following example illustrates the theorems above. The computations use the addition properties (i) $O(h^p) + O(h^p) = O(h^p)$, (ii) $O(h^p) + O(h^q) = O(h^r)$, where $r = \min\{p, q\}$, and the multiplicative property (iii) $O(h^p)O(h^q) = O(h^s)$, where $s = p + q$.

Example 1.22. Consider the Taylor polynomial expansions

$$e^h = 1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + O(h^4) \quad \text{and} \quad \cos(h) = 1 - \frac{h^2}{2!} + \frac{h^4}{4!} + O(h^6).$$

Determine the order of approximation for their sum and product.

For the sum we have

$$\begin{aligned} e^h + \cos(h) &= 1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + O(h^4) + 1 - \frac{h^2}{2!} + \frac{h^4}{4!} + O(h^6) \\ &= 2 + h + \frac{h^3}{3!} + O(h^4) + \frac{h^4}{4!} + O(h^6). \end{aligned}$$

Since $O(h^4) + \frac{h^4}{4!} = O(h^4)$ and $O(h^4) + O(h^6) = O(h^4)$, this reduces to

$$e^h + \cos(h) = 2 + h + \frac{h^3}{3!} + O(h^4),$$

and the order of approximation is $O(h^4)$.

The product is treated similarly:

$$\begin{aligned} e^h \cos(h) &= \left(1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + O(h^4)\right) \left(1 - \frac{h^2}{2!} + \frac{h^4}{4!} + O(h^6)\right) \\ &= \left(1 + h + \frac{h^2}{2!} + \frac{h^3}{3!}\right) \left(1 - \frac{h^2}{2!} + \frac{h^4}{4!}\right) \\ &\quad + \left(1 + h + \frac{h^2}{2!} + \frac{h^3}{3!}\right) O(h^6) + \left(1 - \frac{h^2}{2!} + \frac{h^4}{4!}\right) O(h^4) \\ &\quad + O(h^4)O(h^6) \\ &= 1 + h - \frac{h^3}{3} - \frac{5h^4}{24} - \frac{h^5}{24} + \frac{h^6}{48} + \frac{h^7}{144} \\ &\quad + O(h^6) + O(h^4) + O(h^4)O(h^6). \end{aligned}$$

Since $O(h^4)O(h^6) = O(h^{10})$ and

$$-\frac{5h^4}{24} - \frac{h^5}{24} + \frac{h^6}{48} + \frac{h^7}{144} + O(h^6) + O(h^4) + O(h^{10}) = O(h^4),$$

the preceding equation is simplified to yield

$$e^h \cos(h) = 1 + h - \frac{h^3}{3} + O(h^4),$$

and the order of approximation is $O(h^4)$.

Order of Convergence of a Sequence

Numerical approximations are often arrived at by computing a sequence of approximations that get closer and closer to the answer desired. The definition of big Oh for sequences was given in Definition 1.10, and the definition of order of convergence for a sequence is analogous to that given for functions in Definition 1.11.

Definition 1.12. Suppose that $\lim_{n \rightarrow \infty} x_n = x$ and $\{r_n\}_{n=1}^{\infty}$ is a sequence with $\lim_{n \rightarrow \infty} r_n = 0$. We say that $\{x_n\}_{n=1}^{\infty}$ *converges to x* with the *order of convergence $O(r_n)$* , if there exists a constant $K > 0$ such that

$$\frac{|x_n - x|}{|r_n|} \leq K \quad \text{for } n \text{ sufficiently large.}$$

This is indicated by writing $x_n = x + O(r_n)$, or $x_n \rightarrow x$ with order of convergence $O(r_n)$. ▲

Example 1.23. Let $x_n = \cos(n)/n^2$ and $r_n = 1/n^2$; then $\lim_{n \rightarrow \infty} x_n = 0$ with a rate of convergence $O(\quad)$. This follows immediately from the relation

$$\cos(n) = O(1) \quad \text{for all } n. \quad \blacksquare$$

